6 **(a)** A stationary nucleus of a radioactive nuclide, ${}^{218}_{84}Po$, underwent a chain of decays by the emission of an α and β–particles. The decay is represented by the two equations:

$$ {}^{218}_{84}Po \rightarrow D + \alpha $$
$$ D \rightarrow E + \beta $$

where D and E are the nuclides formed after the decay.

(i) State the nuclear notation for E. [1]

$$ {}^{\cdots\cdots\cdots}_{\cdots\cdots\cdots}E $$

${}^{214}_{83}E$

(ii) Determine the ratio of the kinetic energy of the α–particle to the total kinetic energy of D and α–particle.

By conservation of linear momentum,

$$0 = p_D + p_\alpha$$

$$|p_D| = |p_\alpha| \quad [\text{M1}]$$

$$\frac{\text{KE of } \alpha\text{-particle}}{\text{total KE}}$$

$$= \frac{\dfrac{p_\alpha^2}{m_\alpha}}{\dfrac{p_\alpha^2}{m_\alpha} + \dfrac{p_D^2}{m_D}} \quad [\text{M1}]$$

$$= \frac{\dfrac{1}{m_\alpha}}{\dfrac{1}{m_\alpha} + \dfrac{1}{m_D}}$$

$$= \frac{\dfrac{1}{4u}}{\dfrac{1}{4u} + \dfrac{1}{214u}}$$

$$= 0.982 \quad [\text{A1}]$$

ratio = ……………………… [3]

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(iii) In reality, the β–particles have a range of kinetic energies, instead of a fixed value. Explain why this is so.

Because there are <u>neutrinos emitted</u> together with the beta particles and <u>kinetic energy is shared between the beta particles and neutrino</u>. [A1]

…………………………………………………………………………………………[1]

(iv) A sample of ${}^{218}_{84}Po$ is placed on a weighing balance and a reading of 4.05 g is obtained. After 243 s, the reading drops to 4.02 g.

1. Determine the number of particles ${}^{218}_{84}Po$ in the initial sample.

$$\text{No. of } {}^{218}_{84}Po \text{ particles } \mathrm{N}_{218} = \frac{4.05\times10^{-3}}{218u} = 1.12\times10^{22} \quad [\mathrm{A1}]$$

number of particles = ………………….. [1]

2. Show that the total number of particles D and E after 243 s is 4.52 x 10^{21}. [1]

$$\text{Total mass after time } t = (N_{Po} - N_{D+E})m_{Po} + (N_{D+E})m_D$$
$$(1.119\times10^{22} - N_{D+E})218u + (N_{D+E})214u = 4.02\times10^{-3} \quad [\mathrm{M1}]$$
$$(1.119\times10^{22} - N_{D+E})218u + (N_{D+E})214u = 4.02\times10^{-3}$$
$$4.05\times10^{-3} - 4.02\times10^{-3} = (N_{D+E})4u$$
$$N_{D+E} = 4.518\times10^{21}$$

3. Determine the half life of ${}^{218}_{84}Po$.

$$N = N_0\left(\frac{1}{2}\right)^{\frac{t}{t_{\frac{1}{2}}}}$$
$$(1.12\times10^{22} - 4.52\times10^{21}) = 1.12\times10^{22}\left(\frac{1}{2}\right)^{\frac{243}{t_{\frac{1}{2}}}} \quad [\mathrm{M1}]$$
$$t_{\frac{1}{2}} = 326 \text{ s } [\mathrm{A1}]$$

half life = …………………. s [2]

(b) For many unstable parent nuclei, the daughter nuclei is itself radioactive. This may give rise to a radioactive series where there may be ten or more different radioactive daughter products.

A radioactive parent nucleus X has a radioactive daughter nucleus Y and, in turn, this daughter produces a further stable daughter Z. The variation with time t of the percentage number P of the different nuclei X, Y and Z in a radioactive sample is illustrated in Fig. 6.1

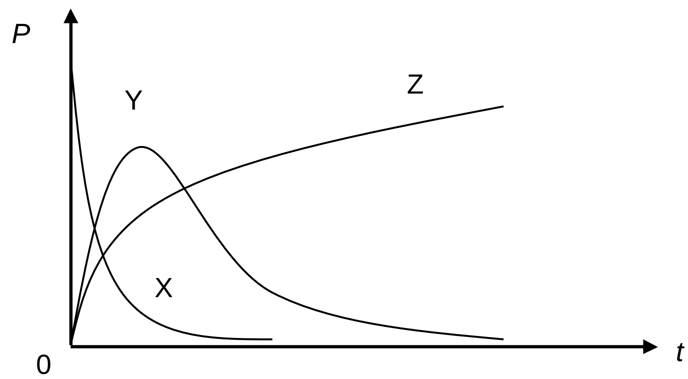

Fig 6.1

(i) X has a half life of 5 hours. The count rate at t = 0 and t = 5 hours were measured. Suggest three possible reasons why the count rate is not exactly halved after 5 hours.

1. ……………………………………………………………………………………………

……………………………………………………………………………………………

1. Existence of background count. [B1]
2. Product Y is also giving off radiation that adds to count rate. [B1]
3. Random nature of radioactive decay. [B1]

2. ……………………………………………………………………………………………

……………………………………………………………………………………………

3. ……………………………………………………………………………………………

……………………………………………………………………………………………[3]

(ii) Explain why graph of Y increases to a maximum and then decreases.

It happens because initially, there are more number of X than Y and hence, Y is being formed faster than it is decaying. [B1] As number of X decreases, there will be a point where Y decays faster than it is being formed. Hence, this will lead to a decrease in number of Y. [B1]

……………………………………………………………………………………………

……………………………………………………………………………………………[2]

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7 Many devices are designed to create a spray of tiny droplets. The effectiveness of these